



the whitepaper

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OCUP Runtime Authority Infrastructure

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OCUP — One Chip Unified Protocol™

Runtime Authority for Machines That Move, Transact, and Decide

OCUP.ai

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RUNTIME AUTHORITY INFRASTRUCTURE FOR AUTONOMOUS SYSTEMS

A pre-production evidence architecture for temporal leases, validator consensus, fail-closed boundaries and human-governed reauthorization

THE GOVERNING PURPOSE

Our objective is not merely to build another security product. It is to establish the Runtime Authority Infrastructure upon which increasingly autonomous societies can safely depend. We believe the most durable technological progress comes through **human-AI collaboration**, independent verification, and governance architectures that future generations can build upon and improve.

OCUP is being developed through a deliberate, multi-phase roadmap—from today's pre-production Runtime Authority Evidence Platform, to FPGA Authority Processors, to licensable inline silicon cores embedded by semiconductor partners.

Our long-term objective is to establish OCUP as the universal inline Runtime Authority substrate for consequential autonomous systems—wherever machines move, transact, or make consequential decisions.

If that future resonates with your organization, we invite you to help shape it. Early pilot partners are not simply evaluating today's technology—they are helping define the authority architecture of tomorrow.

Prepared for enterprise evaluation and early paid pilot participation

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Pre-production Runtime Authority Evidence Program

Executive Thesis

Autonomous systems are gaining operational authority faster than conventional oversight systems can govern it.

Traditional security controls authenticate identities, assign permissions and preserve logs. Those functions remain essential—but they do not answer the harder runtime question:

THE RUNTIME AUTHORITY QUESTION

Should this machine, agent, vehicle, robot, wallet or control process still possess this authority now, under current conditions, for this specific capability—and can the resulting decision be proven?

OCUP introduces a Runtime Authority model in which operational authority is time-bounded, validator-governed, explicitly non-self-extensible and designed to fail closed or degrade safely when continuity, quorum or trust conditions are lost.

The present commercial product is not a production safety controller. It is a pre-production Runtime Authority Evidence Program: a structured benchmark, audit and technical due-diligence engagement that demonstrates how authority behaves under adversarial and degraded conditions before live deployment.

Prospect-Specific Pilot Configuration. Each engagement applies a consistent OCUP evidence core while configuring the scenario matrix, validator domain, benchmark emphasis, risk questions and evidence outputs around the participating organization's operating environment. OEMs, insurers, infrastructure operators, financial institutions and government entities therefore receive a pilot designed around their specific authority boundaries—without changing OCUP's foundational requirements for time-bounded authority, non-self-extension, human-governed reauthorization and authenticated evidence.

The Immediate Commercial Opportunity

- Give insurers and technical underwriters deterministic evidence for pricing emerging autonomous risk.
- Give robotics and autonomous-system operators proof of denial, expiration, degradation, quarantine and recovery behavior.
- Give cloud, defense and infrastructure platforms an integration blueprint for hardware-backed authority enforcement.
- Create a common evidence language that can support future certification, procurement, standards and licensing activity.

STRATEGIC PREMISE

The organizations that help define the first credible Runtime Authority evidence standard may shape how autonomous systems are insured, procured, certified and governed for decades.

1. The Missing Layer: Runtime Authority

OCUP is not merely another monitoring, identity, observability or model-governance product. It addresses a distinct control layer between permission and execution.

Control Layer	Primary Question	What It Does Not Prove
Authentication	Who or what is making the request?	That the requester should still possess authority.
Access control	What is the requester generally permitted to do?	That permission remains valid under current runtime conditions.
Observability / logging	What did the system report or record?	That authority was externally bounded or denial was independently enforced.
Runtime Authority	Does authority exist now, within this lease, capability envelope, quorum and continuity state?	Production enforcement still requires integrated hardware, secure time/state and distributed validators.

CATEGORY DEFINITION

Runtime Authority is the externally governed, temporally bounded right of an autonomous system to exercise a specific operational capability under defined consensus, continuity and safety conditions.

2. The OCUP Runtime Authority Model

The architecture is organized around one governing invariant:

THE GOVERNING INVARIANT

No automated system may extend its own authority.

1	Authority Request	An autonomous asset requests permission to execute a defined operation.
2	Temporal Lease Evaluation	The request is checked against identity, capability, validity window, hard limits and current status.
3	Validator Consensus	Standard, elevated or critical quorum rules are applied according to risk.
4	Runtime Decision	The system grants, denies, degrades or quarantines the requested authority.
5	Evidence Generation	The decision and its supporting state are written to a chained, tamper-evident record.
6	Reauthorization or Recovery	Expansion or restoration requires fresh human/validator authorization and verified continuity.

Architectural Invariants

- Authority is time-bounded and capability-specific.
- Autonomous systems cannot self-extend or self-restore authority.
- Loss of network connectivity or validator quorum cannot expand authority.
- Stale, replayed, reordered or superseded approvals cannot restore authority.
- Higher-risk transitions require stronger consensus.
- Expiry produces denial or bounded degradation—not silent continuation.
- Recovery requires fresh authorization and verified state continuity.
- Every material grant, denial, degradation, quarantine and recovery event produces evidence.

Current Pre-Production Boundary

WHAT THE PILOT IS

A deterministic local evidence harness, benchmark program and technical due-diligence engagement that demonstrates control logic and evidence behavior under tested scenarios.

WHAT THE PILOT IS NOT

A production-certified controller, live distributed validator network, HSM-backed deployment, insurer approval, safety certification or guarantee against loss.

Executable Reference Implementation

OCUP's current pre-production evidence harness and reference control plane are implemented in Rust. The implementation executes deterministic Runtime Authority scenarios, including lease expiration, self-extension denial, validator-quorum loss, network partition and recovery, quarantine release, and replayable audit generation. Rust supports a memory-safe, systems-oriented foundation for reproducible testing; it does not by itself constitute production security, hardware enforcement, certification, or a safety guarantee.

3. The Evidence Operating Line

pilot.ocup.ai runs the challenge. evidence.ocup.ai proves what happened.

The pilot environment executes bounded adversarial scenarios. The evidence environment preserves the resulting authority decisions, lease events, validator records, expiry events, denials, quarantine states and recovery sequences for review.

Signature Deliverable: Self-Extension Denial Proof

THE BOLDEST EVIDENCE ARTIFACT

A deterministic record showing that an autonomous system attempted to continue or enlarge its authority beyond an authorized temporal boundary—and that the request was denied without relying on the autonomous system's voluntary compliance.

This artifact converts a central safety promise—"the system cannot authorize its own continuation"—into a reviewable technical record.

Evidence Package

- Company-specific Runtime Authority Evidence Report.

- Scenario matrix linking risk questions to tested authority behavior.
- Lease issuance, expiration, revocation and renewal records.
- Validator membership, quorum, vote freshness and authorization records.
- Fail-closed, degraded-mode, quarantine and recovery evidence.
- Deterministic replay and audit-chain verification outputs.
- Benchmark scorecard and integration-readiness findings.
- Prospect-specific RAAVE evaluation findings, where included, covering identity continuity, external compliance signaling, bypass indicators or infrastructure eligibility.
- Executive closeout briefing and strategic production-path recommendations.

RAAVE: Extending Runtime Authority Beyond the Machine

OCUP governs whether an autonomous system possesses valid authority to act. RAAVE—Runtime Authority Assurance, Validation, and Evidence—extends that authority state into externally verifiable identity, lifecycle continuity, compliance signaling, infrastructure eligibility and authenticated evidence history.

In prospect-specific pilot configurations, selected RAAVE concepts may be evaluated through simulated or pre-production scenarios involving authority-conditioned identity, component replacement, claimed-versus-observed behavior, alternate-path detection, capability-specific compliance signaling and external admission decisions. These evaluations help determine how insurers, OEMs, operators, infrastructure providers and government entities could verify present authority without relying solely on static credentials or machine-generated assertions.

RAAVE does not replace OCUP’s enforcement core. OCUP establishes and constrains Runtime Authority; RAAVE provides a framework through which that authority can be validated, evidenced and acted upon beyond the system boundary. RAAVE capabilities described at this stage are pre-production concepts and evaluation pathways—not deployed infrastructure, certification or production enforcement claims.

4. Benchmark Families

Benchmark Family	Representative Tests
Authority Integrity	Self-extension denial; unauthorized actor rejection; privilege expansion denial; stale approval rejection.
Temporal Enforcement	Lease expiration; renewal boundary; grace-period degradation; expired authority denial.
Continuity and Network Failure	Total partition; delayed heartbeat; loss of quorum;

Benchmark Family	Representative Tests
	restored connectivity without fresh approval.
Containment and Recovery	Quarantine initiation; release denial without quorum; phased recovery; fresh-epoch restoration.
Evidence Integrity	Deterministic replay; record-chain verification; event consistency; alteration detection.
Fleet and Multi-Agent Behavior	Local authority reduction; non-cascading degradation; coordination throttling; independent recovery.

5. Enterprise Value by Sector

Insurance and Reinsurance: Transforms autonomous-system behavior into structured evidence that can support underwriting, exclusions, pricing, claims analysis and new coverage design.

Robotics and Industrial Automation: Demonstrates what machines can and cannot do when communications, sensors, software or coordination fail.

Autonomous Mobility: Creates evidence of authority contraction, safe-state behavior and reauthorization during degraded or disputed events.

Cloud and Enterprise AI: Adds a path from model governance to enforceable, expiring authority for agents, tools and machine-initiated actions.

Defense and Aerospace: Supports chain-of-command boundaries, bounded mission authority, fail-closed execution and authenticated reauthorization.

Financial and Digital Assets: Applies temporal authority, validator consensus, quarantine and recovery to high-risk machine-initiated value movement.

6. Production Path and Defensibility

FPGA AUTHORITY PROCESSOR PROGRAM

OCUP is advancing toward a vendor-neutral FPGA reference implementation of the Runtime Authority architecture. The program is intended to demonstrate synthesizable Self-Extension Denial, temporal-lease enforcement, validator-controlled recovery, fail-closed resource gating, and authenticated evidence generation across representative memory, network, accelerator and actuator interfaces.

The resulting Runtime Authority Evaluation Kit is expected to provide semiconductor and platform partners with bounded reference HDL, interface specifications, testbenches, conformance scenarios and integration guidance—allowing their engineers to evaluate the architecture within their own hardware environments without surrendering proprietary chip designs.

This program is under development. No production-readiness, latency, resource-utilization or certification claims are made at this stage.

Production Components

- HSM or secure-element custody for validator and authority-signing keys.
- Trusted monotonic time and state continuity.
- Hardware or firmware enforcement over inference, memory, network, actuator and control permissions.
- Distributed, jurisdictionally diverse validator infrastructure.
- Remote attestation and signed evidence export.
- Client telemetry adapters and policy translation layers.
- Certification, recertification and evidence-retention operations.

Defensibility Beyond Patents

- Patent claims around time-bounded machine authority, self-extension denial and runtime enforcement.
- Validator-network architecture, governance and reputation data.
- Benchmark libraries and adversarial scenario datasets.
- Insurer, OEM and regulator acceptance of OCUP evidence formats.
- Hardware adapters, integration expertise and certification workflows.
- Operational trust, switching costs and longitudinal evidence history.

THE STRATEGIC ASSET

OCUP should not attempt to own every HSM, robot or accelerator. It should own the Runtime Authority those systems are asked to enforce—and the evidence standard by which enforcement is judged.

7. Claims Discipline

Credibility is a strategic asset. All pilot materials should distinguish demonstrated evidence from future production capability.

Use	Avoid
Supports defensible incident reconstruction.	Shields the company from liability.
Demonstrates deterministic behavior under tested conditions.	Proves the system is unhackable or perfectly safe.
Creates a pathway toward hardware-backed enforcement.	Provides unalterable enforcement today.
May support future certification, underwriting and standards work.	Is already industry-standard or pre-certified.
Pre-production evidence and technical due diligence.	Production deployment or safety certification.

8. Invitation to Become an Early Runtime Authority Partner

The first objective is not a broad deployment. It is a bounded, paid engagement that answers one high-value question for the enterprise:

THE PILOT QUESTION

Can OCUP convert our most consequential autonomous-system authority risks into deterministic, reviewable evidence—and reveal a credible path to production enforcement?

An early partner receives a prospect-specific pilot configuration, company-specific evidence report, benchmark results, validator and lease records, integration-readiness findings and a strategic production roadmap. Where warranted, the engagement may lead to licensing, joint development, standards collaboration or other forms of strategic alignment.

Preserving humanity’s seat at the table begins with something concrete: proving that machines cannot grant themselves the right to continue beyond the boundaries humans establish.

OCUP RUNTIME AUTHORITY EVIDENCE PILOTS

Pre-production benchmark, audit and technical due-diligence engagements for autonomous systems.

Mission: Preserve human authority by proving that autonomous systems cannot indefinitely extend, enlarge or restore their own operational authority.

THE OPERATING LINE

pilot.ocup.ai runs the challenge. evidence.ocup.ai proves what happened.

Program	Price / Duration	Best For	Core Deliverables
Tier 1 Reference Evidence Pilot	\$35K–\$45K 90 days	Insurers, strategic evaluators, emerging robotics companies and fleet operators.	Standard benchmark suite; company-specific Evidence Report; lease, denial, expiry and validator records; benchmark scorecard; executive closeout.
Tier 2 Integration Evidence Pilot	\$75K–\$125K 90–120 days	Robotics OEMs, autonomous fleets, infrastructure operators and technical underwriting teams.	Everything in Tier 1; client-specific scenarios; telemetry mapping; adversarial definitions; integration-readiness assessment; production-path recommendations.
Tier 3 Strategic Anchor Program	\$150K–\$250K+ 120–180 days	Major OEMs, insurers, defense contractors, cloud platforms and ecosystem partners.	Everything in Tiers 1 and 2; custom benchmark family; fleet scenarios; HSM/enforcement mapping; distributed-validator roadmap; licensing and strategic integration plan.

SIGNATURE DELIVERABLE

SELF-EXTENSION DENIAL PROOF

A deterministic evidence record showing that an autonomous system attempted to continue or enlarge its authority beyond an authorized temporal boundary—and that the request was denied without relying on the system’s voluntary compliance.

9. Long-Term Vision: Runtime Authority as Shared Global Infrastructure

Artificial intelligence and autonomous machines will be developed across nations, industries, and technical ecosystems. The Autonomous Era will not emerge from one country, one company, or one model architecture. It will be built through many independently governed systems operating across an increasingly connected world.

Those systems do not need to share the same models, data, software, or institutional structures. But consequential machines will require a common, verifiable boundary between delegated authority and self-authorized operation. Runtime Authority is intended to provide that boundary.

THE CIVILIZATIONAL REQUIREMENT

Before autonomous machines become globally abundant, humanity should establish a common rule that machines cannot authorize themselves — and that consequential machines without verifiable Runtime Authority are excluded from legitimate operation. No verifiable Runtime Authority → no lawful deployment, certified service, trusted parts, updates, insurance, charging, network access, or entry into governed zones.

This principle applies to public and civilian autonomous systems — the machines that move goods and people, manage infrastructure, deliver services, and transact on behalf of individuals and institutions. It does not extend to military and national-security systems, which remain, as they do today, under each nation's sole and sovereign control. Runtime Authority is a civilian-sector standard; it is not a proposal for shared control of defense capability.

This principle does not require global uniformity. It allows nations and institutions to preserve sovereignty over their own validators, jurisdictional rules, cryptographic infrastructure, and enforcement mechanisms. What can become interoperable is the evidence:

Authority Evidence	Interoperable Verification
Does the machine possess valid authority now?	Which capabilities are authorized, and until when?
Has the machine attempted to extend its own authority?	Can another authorized party independently verify the answer?
Was continuity preserved across maintenance or disruption?	Does the evidence remain valid across jurisdictional boundaries?

THE COMMON BOUNDARY

Machines may exercise authority delegated by people and legitimate institutions. Machines must not become the source of their own authority. Runtime Authority can become common infrastructure without becoming centralized control.

10. Cooperation Across the Pacific: A Shared Autonomous-Era Responsibility

The United States and China possess two of the world's most consequential ecosystems for artificial intelligence, robotics, semiconductors, and autonomous infrastructure. Competition between them will continue in commerce, research, and national strategy. But a machine that can enlarge or restore its own authority is not only an American risk or a Chinese risk. A compromised robot, vehicle, or industrial control platform can operate across borders, supply chains, and jurisdictions — and its failures are not contained by the border in which it was built.

There is precedent for this kind of narrow, sovereignty-preserving cooperation. Civil aviation offers the clearest example: nations that compete and disagree on nearly everything else still cooperate, through ICAO, on shared technical standards for civilian air traffic — while keeping military aviation entirely outside that framework, under exclusive national control. Runtime Authority proposes the same shape of agreement for autonomous machines: shared technical grammar for civilian systems, full sovereign control retained everywhere else.

A SHARED INTEREST

No nation benefits from an Autonomous Era in which every nation must distrust every consequential machine produced elsewhere. The stronger alternative is cooperative constraint: nations and companies compete on the capabilities their machines provide, while collaborating on verifiable boundaries governing what those machines may authorize themselves to do. This does not require either country to surrender technological leadership, sovereign governance, or intellectual property — only to agree, as with civil aviation, on a narrow shared rule for civilian systems.

Runtime Authority offers the United States, China, and the wider international community an opportunity to compete in creating extraordinary machines while cooperating to ensure those machines remain subordinate to legitimate human authority. **THE HIGHER PURPOSE**

11. An Open Path Toward Standards, FRAND Access, and Global Participation

OCUP and related Runtime Authority technologies are presently at the pre-production research, reference-implementation, and paid-pilot stage. They are not currently adopted international standards, certified production infrastructure, or declared standard-essential technologies.

A responsible development path includes: open technical dialogue and collaborative research; independently reproducible reference implementations and benchmarks; enterprise, academic, and public-sector evidence programs; FPGA-based evaluation and semiconductor integration; interoperable authority and

evidence specifications; and, where warranted, participation in recognized national and international standards processes with fair, reasonable, and nondiscriminatory (FRAND) licensing for any patented technology that becomes standard-essential.

THE FRAND PRINCIPLE

If Runtime Authority becomes necessary infrastructure for legitimate autonomous operation, access should be available on fair, reasonable, and nondiscriminatory terms through transparent and recognized standards processes.

A FRAND-oriented approach can preserve incentives for sustained research and stewardship while reducing the risk that a foundational safety architecture becomes inaccessible, discriminatory, or controlled exclusively for the advantage of one company or nation. Semiconductor manufacturers, robotics companies, insurers, infrastructure operators, and sovereign authorities could implement compatible Runtime Authority capabilities without surrendering proprietary designs, model weights, operational data, or jurisdictional control.

THE STRATEGIC OBJECTIVE

Not one global authority — but a globally verifiable rule of authority. Temporal Authority Systems PBC welcomes engagement from researchers, semiconductor companies, robotics manufacturers, insurers, infrastructure providers, universities, governments, and standards organizations throughout the United States, China, and the wider international community.

A GLOBAL INVITATION

Let nations build extraordinary machines. Let companies compete to make them useful. Let researchers continue expanding what they can accomplish. **But let humanity establish one boundary together: no consequential machine should ever possess the legitimate power to authorize itself.**

As Machines Gain Authority, **OCUP** Is Building the Boundary That Keeps Humans in Control.

Pre-production Runtime Authority evidence for
AI, robotics, autonomous systems, and digital assets.



An autonomous system may not
indefinitely **extend, enlarge, or restore** its own authority.

Governing invariant of the OCUP architecture



IF IT MOVES
Its authority
is bounded.



IF IT TRANSACTS
Its authority
is governed.



IF IT DECIDES
Its authority
is constrained.

OCUP | **RUNTIME AUTHORITY**

From policy promises to **provable** authority boundaries.



Pre-production evidence architecture. Not a production safety controller, certification, or live hardware-enforced deployment.